UJWAL MAHAJAN

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AI/GenAI Engineer | Agentic AI & Multi-Agent Systems | Full-Stack AI Engineer

SUMMARY

AI/GenAI engineer focused on agentic AI and multi-agent systems — building orchestration-heavy platforms rather than API wrappers, with emphasis on failure modes, memory, and inter-agent coordination. Currently building a large language model from scratch (transformer blocks, attention, gradient updates) to understand LLM internals beyond the API layer. Portfolio: ujwalmahajan.vercel.app

EDUCATION

B.Tech in Electronics & Telecommunication Engineering *Jan 2023 – Present*

Shri Guru Gobind Singhji Institute of Engineering & Technology, Nanded *Maharashtra, India*

EXPERIENCE

AI Engineering Intern | BluQ *3 Months*

- Engineered full-stack features end-to-end (React, Fastify, Supabase) and architected 2 real-time communication pipelines (WhatsApp + email), automating daily progress capture across 100+ concurrent contractor–PM interactions.
- Architected the cross-platform iOS/Android app blueprint (React Native/Expo, Supabase Auth, BFF API strategy) while independently driving international market research across 4 countries — sourcing and validating 21,000+ firms across 10+ industry directories.
- Drove 5+ critical infrastructure decisions directly with founders, spanning AI-based delay-prediction modeling, per-user rate-limiting design under load (300 req/min), and scalability planning for a shared Fastify/Supabase backend.

PROJECTS

CodeSensei – AI Code Intelligence Platform *Dec 2025*

- Architected a multi-agent AI orchestration framework — 5 specialized agents (structure, behavior, risk, execution, impact) with strict JSON-structured outputs and synthesizer-based merge logic — for automated GitHub repository analysis.
- Engineered an end-to-end code intelligence pipeline processing 5–10K LOC codebases (50–100 files) in 30–90s, delivering sub-10-second semantic query responses validated against known dependencies and execution paths.
- Built an interactive D3.js knowledge-graph visualization and cinematic execution tracer for security and risk analysis, mapping runtime behavior and code dependencies; integrated React/TypeScript frontend with a Node.js backend proxy.

Stack: React, TypeScript, Node.js, Gemini API, D3.js, Multi-Agent Orchestration | GitHub | Deployed Website

TalkToWeb – GenAI Browser Automation Agent *Sep 2025*

- Engineered an agentic browser automation system executing natural-language commands for multi-step workflows (navigation, form-filling, web scraping) with high functional correctness across common tasks.
- Designed an intent parser translating user commands into executable action plans; built a modular "skills" framework and voice interface to extend browser automation workflows.
- Orchestrated a LangChain + Gemini 1.5 Flash pipeline with Playwright for headless browser control, natural-language task automation, and real-time execution.
- **Stack:** LangChain, Gemini API, Playwright, Python, FastAPI

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C, C++

AI, LLM & ML: LangChain, LangSmith, OpenAI API, Gemini API, Prompt Engineering, Multi-Agent Orchestration, Agentic AI, RAG Systems, Semantic Search, PyTorch, Scikit-learn, Pandas, NumPy

Frontend: React, React Native/Expo, D3.js

Backend: Node.js, FastAPI, Fastify, REST APIs

Databases & Vector Stores: FAISS, ChromaDB, Pinecone, Supabase, Neo4j

Dev Tools & DevOps: Git, Docker, Playwright, Streamlit, MATLAB

CERTIFICATIONS

- Building a Large Language Model from Scratch – Vizuara.ai
- Prompt Engineering for Developers – DeepLearning.AI

LEADERSHIP & TECHNICAL EXPERIENCE

Lead Organizer, Rmageddon 2026 | RNXG Robotics Club *Sep 2025 – Mar 2026*

- Leading operations for a national-level robotics competition drawing teams from across India — owning event logistics, sponsorships, and technical coordination as a core committee member.

Finalist, International Rover Challenge (IRC) 2025 *2024 – 2025*

- Programmed autonomous rover systems for navigation, object detection, and task execution on simulated Mars terrain; built control algorithms and sensor-integration logic.